Transposition violates the proposed three-qubit cb ratio

Disproof of Conjecture 00000000998

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Abstract

The observable algebra of three qubits is $M_8(\mathbb{C})$. Its transpose map τ has ordinary induced norm one, while its completely bounded norm is at least eight. We give an explicit matrix amplification witnessing this bound and a uniform upper bound of 64, ensuring that the cb norm is a genuine finite supremum. Thus $\sqrt{2}$ is not even an upper bound for the stated ratios. Lean verifies actual Euclidean operator norms, all amplification sizes and the concrete witness.

The norms and an actual completely bounded map

Equip $M_8(\mathbb{C}) = B(\mathbb{C}^8)$ with the matrix operator norm induced by the Euclidean norm. For a complex-linear map $T : M_8 \to M_8$, its ordinary norm is

$$\|T\| = \sup_{A \neq 0} \frac{\|T(A)\|}{\|A\|}.$$

Its amplification $T^{(k)} = \mathrm{id}_{M_k} \otimes T$ acts on each 8 by 8 block of $M_{8k}(\mathbb{C})$. All amplified matrix norms are also Euclidean operator norms. The completely bounded norm is

$$\|T\|_{\mathrm{cb}} = \sup_{k \geq 1} \|T^{(k)}\| = \sup_{k \geq 1,\ X \neq 0} \frac{\|T^{(k)}(X)\|}{\|X\|}.$$

These are the usual operator-space definitions [2, p. 22]. The source imposes no restriction to completely positive maps.

Consider $\tau(A) = A^{\mathsf{T}}$. Transposition preserves the matrix operator norm: if J denotes coordinatewise complex conjugation, then $A^{\mathsf{T}} = JA^*J$, and J is a conjugate-linear isometry. Hence $\|\tau\| = 1$.

Let E_{ij} be the matrix units, $1 \leq i, j \leq 8$. For every k and every $X \in M_{8k}$, put $B_{ij} = I_k \otimes E_{ij}$. Direct multiplication gives

$$\tau^{(k)}(X) = \sum_{i,j=1}^{8} B_{ij} X B_{ij}, \qquad \|B_{ij}\| \leq 1.$$

The factors on the two sides are the same matrix unit. The triangle inequality and submultiplicativity imply

$$\|\tau^{(k)}(X)\| \leq 64\|X\| \quad \text{for every } k, X.$$

Thus τ is completely bounded and $\|\tau\|_{\mathrm{cb}} \leq 64$. This rough bound suffices; no sharp upper bound is needed.

A concrete amplification with norm at least eight

Index $\mathbb{C}^{64}$ by pairs $(a,b) \in \{1,\dots,8\}^2$. Define the swap matrix and diagonal vector by

$$F_{(a,b),(c,d)} = \delta_{a,d}\delta_{b,c}, \qquad \Omega_{(a,b)} = \delta_{a,b}.$$

The matrix F interchanges the two tensor factors. It satisfies $F^* = F$ and $F^2 = I_{64}$, so $\|F\| = 1$. Also $\Omega \neq 0$ and $\|\Omega\|^2 = 8$.

Applying transposition within each block gives

$$Q := \tau^{(8)}(F), \qquad Q_{(a,b),(c,d)} = \delta_{a,b}\delta_{c,d}.$$

In particular $Q = \Omega\Omega^*$ and $Q\Omega = 8\Omega$. The operator norm inequality on this nonzero vector yields

$$8\|\Omega\| = \|Q\Omega\| \leq \|Q\|\|\Omega\|, \qquad \|Q\| \geq 8.$$

Consequently

$$\frac{\|\tau\|_{\mathrm{cb}}}{\|\tau\|} = \|\tau\|_{\mathrm{cb}} \geq \frac{\|\tau^{(8)}(F)\|}{\|F\|} \geq 8 > \sqrt{2}.$$

Theorem 1. *Over the three-qubit observable algebra, $\sqrt{2}$ is not an upper bound for the cb-norm to ordinary-norm ratios of nonzero completely bounded complex-linear maps. In particular it is not their supremum.*

Proof. The map τ is admissible: its ordinary norm is one and its cb norm is finite. Its ratio is at least eight, strictly larger than the claimed value. A supremum must be an upper bound, so the conjecture is false. □

This is a standard transpose-amplification phenomenon, not a novelty claim. The disproof needs only the certified bounds $8 \leq \|\tau\|_{\mathrm{cb}} \leq 64$ and $\|\tau\| = 1$. It does not claim to formalize the sharp value of the global supremum or the sharp cb norm of transposition. The opposite algebra used in the ordinary-norm argument does not change the target operator-space structure of the amplified map: every amplification above acts into the original $M_{8k}(\mathbb{C})$.

Formalization and reproducibility

The type `Obs` is the actual complex matrix algebra on `Fin 8`.

The active Mathlib `Matrix.L2OpNorm` structure is the operator norm from the associated Euclidean continuous linear map. The map `transposeMap` is a genuine complex continuous linear map. Its ordinary norm-one theorem uses transposition as a star-algebra equivalence to the opposite algebra solely to prove its level-one isometry.

The function `amplify` explicitly applies that map to each block, using matrix indices `Fin k` times `Fin 8`. The all-k bound is proved by the matrix-unit identity above. The swap, its unitarity, the nonzero Euclidean vector Ω and the actual matrix-vector eigen-equation are checked in Lean. They prove the lower bound using the actual Euclidean matrix operator norm.

The definition `cbNorm` is the supremum of all amplification norm ratios. The zero input and the zero-dimensional amplification contribute only zero. The witness makes the set nonempty and the 64 bound makes it bounded above, so no empty-set or unbounded-set default value is used. The set `cbRatios` includes only nonzero maps with bounded amplification ratios. The theorem `not_claimed_supremum` directly negates an `IsLUB` certificate at $\sqrt{2}$; the stronger theorem `not_upperBound` also holds.

Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From the submitted `lean/` directory:

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The complete project is freshly built and the strict check treats warnings as errors. The axiom audit contains only standard foundational axioms. No proof placeholders, custom axioms or external computation oracles are used.

The supplementary `verify.py` checks the swap and rank-one identities exactly with integer arithmetic for the matrix algebras M_2 and M_8. It does not estimate norms numerically or supply a Lean oracle. Run `python verify.py` and `tectonic main.tex`. The exact source, provenance, build logs and artifact hashes are included. An author agent supplied a self-review; the parent separately reviewed the complete proof and final artifacts. No external independent review is claimed.

References

[1] The Last Math Competition, Conjecture 00000000998. Exact bilingual source and provenance included.

[2] V. I. Paulsen, *Operator Algebra Methods in QIT*, Winter 2020, p. 22. Author-hosted lecture notes. The standard definitions and transpose example are background; all estimates used here are proved explicitly above.